

Social Engineering & Security Awareness

Topics:

What is Social Engineering?

Types: Phishing, Pretexting, Baiting, Tailgating, etc.

Real-world case studies of major social engineering attacks.

Tasks:

Research and summarize one real-world social engineering attack (e.g., Twitter Bitcoin Scam, RSA Phishing Attack).

Write a short article/post explaining the importance of cybersecurity awareness.

Phishing Attacks & Email Security

Topics:

How phishing attacks work.

Identifying phishing emails.

Email security best practices (SPF, DKIM, DMARC).

Tasks:

Find and analyze a real phishing email (safely) or use phishing test websites

Set up SPF, DKIM, and DMARC on a test domain (if possible).

Social Engineering & Security Awareness

What is Social Engineering?

Social engineering is a manipulation technique that exploits human psychology to gain access to confidential information or systems. Attackers often use deception, persuasion, or impersonation to trick individuals into revealing sensitive data or performing actions that compromise security.

Types of Social Engineering Attacks:

1. **Phishing:** Fraudulent emails or messages designed to steal sensitive information.
2. **Pretexting:** Creating a fabricated scenario to obtain information or access.
3. **Baiting:** Offering something enticing (e.g., free software) to lure victims into compromising their systems.

4. **Tailgating:** Gaining physical access to restricted areas by following authorized personnel.

Real-World Case Study: Twitter Bitcoin Scam

In July 2020, hackers executed a social engineering attack targeting Twitter employees. They gained access to internal tools and compromised 130 high-profile accounts, including those of Elon Musk, Barack Obama, and Apple. The attackers posted tweets promoting a Bitcoin scam, promising to double any cryptocurrency sent to a specific wallet. The scam resulted in over \$100,000 in Bitcoin transactions.

Importance of Cybersecurity Awareness

Cybersecurity awareness is crucial in today's digital age, where cyber threats are increasingly sophisticated. It involves educating individuals and organizations about potential risks and best practices to mitigate them. Awareness helps prevent breaches caused by human error, which accounts for a significant percentage of security incidents.

Phishing Attacks & Email Security

How Phishing Attacks Work:

Phishing attacks use deceptive emails or messages to trick individuals into revealing sensitive information, such as login credentials or financial details. Attackers often impersonate trusted entities and create a sense of urgency to manipulate victims.

Identifying Phishing Emails:

1. Generic greetings (e.g., "Dear Customer").
2. Misspellings or grammatical errors.
3. Suspicious links or attachments.
4. Requests for sensitive information.

Email Security Best Practices:

1. **SPF (Sender Policy Framework):** Ensures emails are sent from authorized servers.
2. **DKIM (DomainKeys Identified Mail):** Verifies the authenticity of email messages using cryptographic signatures¹⁴.
3. **DMARC (Domain-based Message Authentication, Reporting, and Conformance):** Specifies actions for handling emails that fail SPF or DKIM checks.